

What is the clinical significance of TPM3-NTRK1 Fusion in Lung Carcinoma with Larotrectinib in CIViC?

Understood. I'll search for the clinical significance of the TPM3-NTRK1 fusion in lung carcinoma, specifically regarding larotrectinib, within the CIViC database or other relevant up-to-date sources. I'll provide you with any relevant findings and cited information.

Worked for 12 minutes >

In the CIViC database, the TPM3-NTRK1 gene fusion in lung carcinoma is classified as a **predictive** biomarker for larotrectinib therapy. The evidence comes from a case-study-level (AMP/ASCO/CAP level C) assertion where larotrectinib produced durable responses in patients with TRK-fusion-positive lung cancers [civcdb.org](#). In the underlying study, 20 patients with TRK-fusion-positive lung cancers were treated with larotrectinib; 19 had adenocarcinoma and one had neuroendocrine carcinoma. Fifteen patients were evaluable and the objective response rate was 73 % (95 % CI 45–92), with one complete response, 10 partial responses, three cases of stable disease and one progressive disease [pmc.ncbi.nlm.nih.gov](#). Median duration of response (DoR), progression-free survival (PFS) and overall survival (OS) were 33.9, 35.4 and 40.7 months, respectively [pmc.ncbi.nlm.nih.gov](#).

The TPM3-NTRK1-specific data in CIViC describe a 46-year-old male with stage IV non-small-cell lung carcinoma who had progressed on platinum-based chemotherapy and the PD-1 inhibitor dostarlimab. Molecular profiling identified a **TPM3-NTRK1 fusion**; larotrectinib therapy produced a brisk partial response at 1.8 months and maintained disease control for 39 months [pmc.ncbi.nlm.nih.gov](#). This evidence underlies CIViC's assertion that the TPM3-NTRK1 fusion is a **sensitivity/response** marker for larotrectinib (direction "supports") and that the variant's origin is **somatic** [civcdb.org](#). Thus, CIViC concludes that lung carcinoma patients whose tumors harbor a TPM3-NTRK1 fusion may respond favorably to larotrectinib, and testing for this fusion can guide targeted therapy decisions.

Sources



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